

Lakshya JEE 2.0 2025

Maths

DPP: 5

Basics Mathematics

Q1 The sum of three numbers in A.P. is 27 and the sum of their square is 293. Then numbers are

- (A) 5, 10, 15
(B) 4, 9, 14
(C) 3, 8, 13
(D) 6, 11, 16

Q2 If $\log_5 2, \log_5(2^x + 1)$ and $\log_5(2^x + 5/2)$ are in A.P. then x is equal to

- (A) 1
(B) 2
(C) 3
(D) 4

Q3 If 100 times the 100^{th} terms of an A.P. with non-zero common difference equals the 50 times its 50^{th} term, then the 150^{th} term of this A.P. is

- (A) 150
(B) zero
(C) -150
(D) 150 times its 50^{th} term

Q4 If the 5^{th} term of an A.P. is 9 and 9^{th} term is 5, then which of the following term is unity

- (A) 12^{th}
(B) 13^{th}
(C) 14^{th}
(D) 15^{th}

Q5 If $\log x, \log \sqrt{6-2x}, \log(x-1)$ are in AP then the number of possible values of x is/are

- (A) 0
(B) 1
(C) 2
(D) 3

Q6

For an AP, $T_{10} = \frac{1}{20}$ and $T_{20} = \frac{1}{10}$ then

$$S_{200} =$$

- (A) $\frac{361}{2}$
(B) $\frac{201}{2}$
(C) $\frac{101}{2}$
(D) 50

Q7 $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \dots$ to ∞ is equal to

- (A) $\frac{1}{2}$
(B) 1
(C) 2
(D) 3

Q8 If sum of an infinite geometric series is $\frac{4}{3}$ and its 1^{st} term is $\frac{3}{4}$, then its common ratio is

- (A) $\frac{7}{16}$
(B) $\frac{9}{16}$
(C) $\frac{1}{9}$
(D) $\frac{7}{9}$

Q9 $\sum_{n=1}^{\infty} 3\left(\frac{1}{2}\right)^{n-1} =$

- (A) 3
(B) $\frac{1}{2}$
(C) 6
(D) 12

Q10 The third term of a G.P. is 3. The product of its first five terms is

- (A) 81
(B) 243
(C) 729
(D) 343

Q11 The fifth terms of a GP. is 32 and common ratio is 2, then the sum of first 14 terms of the G.P. is

- (A) 16388
(B) 32667
(C) 32766
(D) 64342



Q12 If the AM of the roots of quadratic equation is $\frac{8}{5}$ and AM of their reciprocal is $\frac{8}{7}$ then the equation is

- (A) $5x^2 - 16x + 7 = 0$
 (B) $7x^2 - 16x + 5 = 0$
 (C) $7x^2 - 16x + 8 = 0$
 (D) $3x^2 - 12x + 7 = 0$

Q13 $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{r(r+1)} =$

- (A) 0 (B) 1
 (C) 2 (D) Does not exist

Q14 If the sum to infinity of the series $1 + 2r + 3r^2 + 4r^3 + \dots$ is $9/4$, value of r is

- (A) $1/2$ (B) $1/3$
 (C) $1/4$ (D) None of these

Q15 $\frac{1^3+2^3+3^3+4^3+\dots+12^3}{1^2+2^2+3^2+4^2+\dots+12^2} =$

- (A) $\frac{234}{25}$
 (B) $\frac{243}{35}$
 (C) $\frac{263}{27}$
 (D) None of these

Q16 If $S = \sum_{r=1}^{80} \frac{r}{(r^4+r^2+1)}$, then the value of $\frac{6481S}{1000}$ is

- (A) 3.2
 (B) 3.24
 (C) 3.26
 (D) 3.28

Q17 The minimum value of the sum of the numbers a^{-5}, a^{-4}, a^9 is

- (A) 1 (B) 2
 (C) 3 (D) 5

Q18 If a, b, c are positive numbers and $abc = 1$ then minimum value of $a + b + c$ is

Q19 The minimum value of $4^x + 4^{1-x}, x \in R$ is

- (A) 1 (B) 2
 (C) 3 (D) 4



Answer Key

Q1 (B)
Q2 (A)
Q3 (B)
Q4 (B)
Q5 (B)
Q6 (B)
Q7 (B)
Q8 (A)
Q9 (C)
Q10 (B)

Q11 (C)
Q12 (A)
Q13 (B)
Q14 (B)
Q15 (A)
Q16 (B)
Q17 (C)
Q18 3
Q19 (D)



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